Elements, PCA and k-means

Feature selection with PCA, and clustering compared with the periodic table

Q1 · Which of the 8 element features matter most?

Q2 · Do k-means clusters match places in the periodic table?

Elements analysed

99

of 118 (19 superheavy removed)

Most important feature

MeltingPoint

PCA method of Song et al. (2010)

Table neighbours in the same cluster

73%

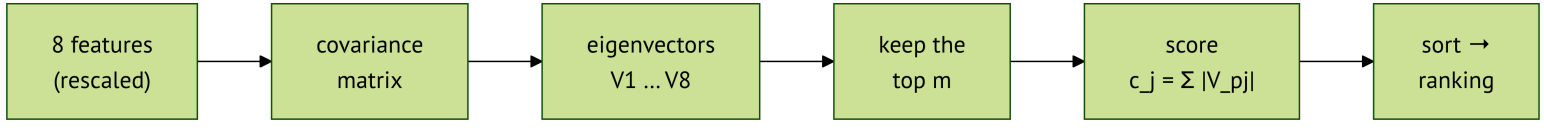
20% expected by chance

The data

118 elements · 8 numeric features: atomic mass, electronegativity, atomic radius, ionization energy, electron affinity, melting point, boiling point, density.

Problem	What we do
`YearDiscovered` = "Ancient"	flag `DiscoveredAncient`; year left empty
`StandardState` = "Expected to be a Solid"	keep "Solid"; flag `StatePredicted`
Elements 100–118 have at most 4 of 8 features	remove them (19 elements)
No electron affinity for noble gases, group 2, Zn, Cd, Hg, N, Mn	set to 0 (the atom does not hold an extra electron)
42 other blanks	fill from the 5 most similar elements (KNN), checked against the **mendeleev** library
Units differ by up to 10^6	rescale each feature to mean 0, spread 1

Q1 · The method of Song, Guo & Mei (2010)

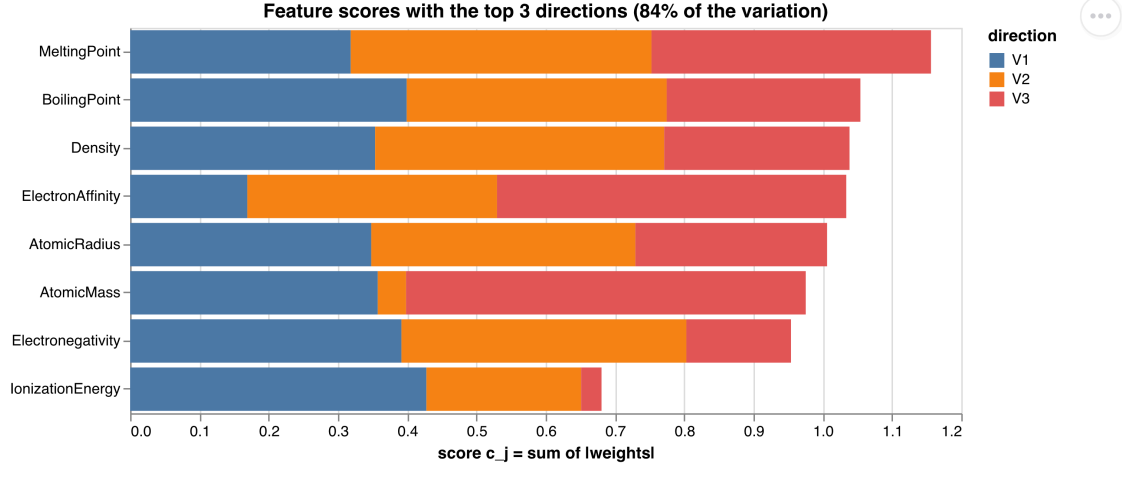


PCA finds **directions** in the data. Each direction is a mix of the 8 features. A feature with **small weights** in the important directions adds little, so it ranks low.

$$c_j = \sum_{p=1}^{m} |V_{pj}| \qquad m = \text{number of directions with eigenvalue} > 1$$

Q1 · The ranking

Top **3** directions = **84%** of the variation. Each bar is a feature's score; the colours show what each direction adds.



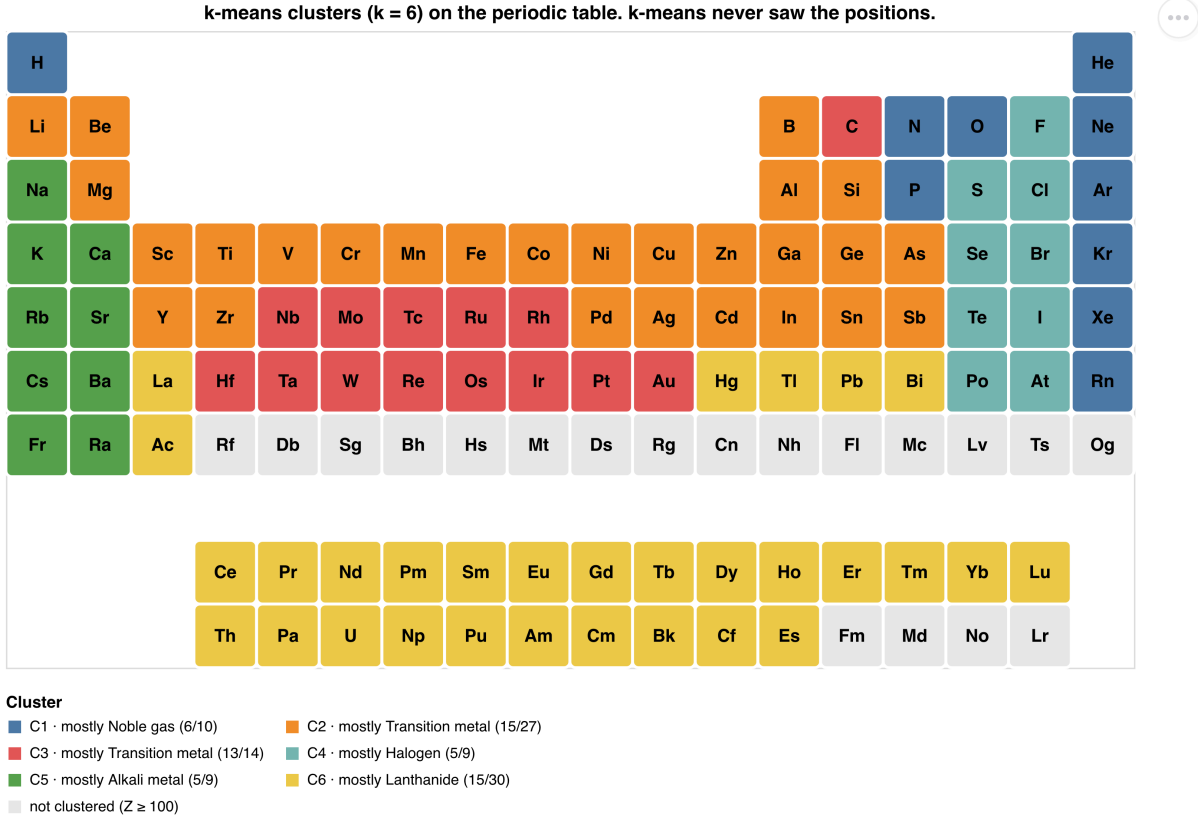
Q1 · Answer

MeltingPoint > BoilingPoint > Density > ElectronAffinity > AtomicRadius > AtomicMass > Electronegativity > IonizationEnergy

- **Stable ends.** Over 1000 bootstrap repeats, MeltingPoint stays in the top 3 in 98% of cases; IonizationEnergy stays in the bottom 3 in 100%.
- **Unstable middle.** The middle ranks change with m and with small changes in the data.
- **Rescale first.** Without rescaling, the kelvin temperatures take 96% of the variation and decide the ranking.
- **Limit.** The method does not see repeated information. Melting and boiling point move together, so both rank high. The top 3 keep $R^2 = 0.53$ of all features; 3 random features keep 0.67.

Q2 · k-means clusters (k = 6) on the periodic table

k-means saw only the 8 features, **never** the positions. Hover an element for its data.



Q2 · Are table neighbours in the same cluster?

Neighbours touch in the table: side by side, or one above the other. We shuffle the labels 5000 times to see what chance gives.

Neighbours in the same cluster	Expected by chance	p-value
73%	**20%**	**0.0002**
154 pairs	5000 shuffles	no shuffle came close

- Down a group: **72%** vs 20% by chance. Along a period: **75%** vs 20%.
- Match (ARI, 0 = chance, 1 = perfect): group 0.42, family 0.39, period 0.25.
- Same result with other ways to fill the blanks, and with no filled values at all.

Q2 · Answer: the clusters are areas of the table

Cluster	Area of the table	Typical features
1	Noble gases + H, N, O, P	gases, highest ionization energy
2	Light metals and semi-metals (Li–Mg, Sc–Zn, B–Sb)	middle values
3	Centre of the heavy transition metals + C	highest melting point and density
4	Halogens and chalcogens	highest electron affinity
5	Groups 1–2 (Na–Fr, Ca–Ra)	lowest electronegativity, largest atoms
6	Lanthanides, actinides + Hg, Tl, Pb, Bi	heavy, large

Exceptions make chemical sense: H is a gas, so it joins the noble gases. C has the highest melting point, so it joins the hard metals.

Summary

Q1. PCA (Song et al.) ranks **melting point** first and **ionization energy** last. Both ends are stable. The method finds the features that drive the main patterns, but it keeps features that repeat each other.

Q2. Yes: k-means groups follow the periodic table. Neighbours share a cluster 3–4× more often than chance. The clusters follow groups and blocks more than periods.

Full analysis, code and interactive charts: `notebook.py`.